

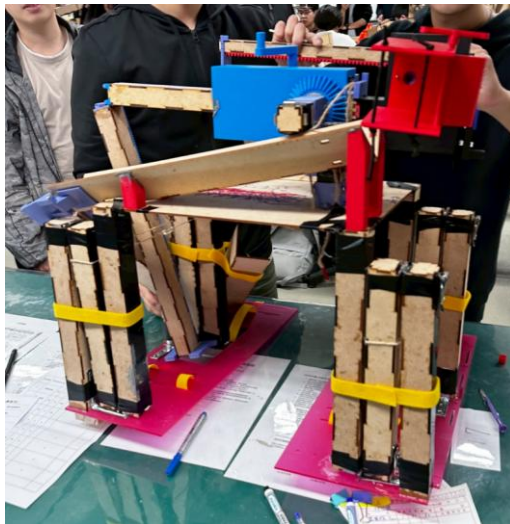
Department of Mechanical Engineering, National Taiwan University

Special topic on mechanical design principles in the first semester of the

113 academic year

Final written report

Title: Polar coordinate positioning boom system



Student number	name	Work content	investment percentage
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Summary

This topic completes a set of polar coordinate positioning crane systems that meet the specifications. Its function is to complete the task of picking up the ball from platform A and transporting it to platform B within a limited time without direct contact with the ball and the storage platform. The system only allows rotation input, and must meet multiple restrictions such as platform height difference, horizontal distance, single ball retrieval and storage volume.

The overall design adopts the polar coordinate positioning concept of a tower crane as the main structure, and achieves spatial positioning through rotation (θ), radial displacement (r) and lifting (z) degrees of freedom; the ball retrieval mechanism refers to the passive guidance and one-way anti-drop design of the table tennis ball picker, so that the balls can passively enter the ball retrieval

container without manual sorting; the transportation part is based on the slope slide, using gravity to complete the stable transmission of the balls, reducing the risk of falling and energy loss.

The system is integrated by a ball retrieval module, a polar positioning module and a gravity conveying module, and converts rotational input into linear and angular motion through transmission elements such as gears, racks, bevel gears and reels. In order to improve operational stability and safety, the Z-axis lifting system introduces a ratchet-pawl self-locking mechanism to prevent gravity from falling back. The R-axis uses multi-gear parallel drive to share the load and reduce the risk of gear failure. The overall structure is also designed with a foldable mechanism to comply with storage size restrictions.

Theoretical calculations and experimental verification results show that all key components of this system have sufficient safety factors under actual operation, and the overall mechanism can stably complete the ball retrieval, positioning and transportation processes, successfully achieving the engineering design goals of this topic.

Chapter 1, device introduction and design goals: design review, introduction to the principle of action, and design goals of this topic

1.1 Device introduction and design goals

The core task of this topic is a "pickup-putting" system that can be installed outside the existing test environment: within the specified time, the organization completes picking up the ball from platform A, transporting it and placing it on platform B, and the number of effective balls left in the B box is used as the score.

Geometric limitations:

A and B are both storage boxes of $60\text{ cm} \times 60\text{ cm}$ and 10 cm deep; the bottom of A is 106 cm from the ground, and the bottom of B is 126 cm from the ground. The horizontal distance between them is 120 cm, and there is a connecting bridge in the middle for fixation. The ball diameter is 7 cm and comes in two weights (approximately 58 g and 111 g).

Operational restrictions:

During the test, the A/B platform or the ball must not be touched directly; the device can touch the box but must not cause it to shift or deform; and the environment must not be damaged.

Functional limitations:

Only one ball can be carried at a time, and the previous ball cannot be removed until the previous ball has passed the center line; the horizontal distance between balls must not be less than 60 cm.

Input form restrictions:

The system input can only be "rotation" (no limit on the number), and the input shaft can only withstand torque and is not allowed to have non-torque external forces; only the member responsible for the input is allowed to touch during operation.

1.2 Design Review: Thinking from the Design of Conveyor Mechanisms in Daily Life

(1). Tower Crane: (Polar coordinate positioning system)



Figure 1. Tower crane[1]

A. Design review and working principle

The tower crane is essentially a cylindrical coordinate system, which is very suitable to be used as the main mechanism for transporting balls in this topic. The core actions are: slewing (angle θ), trolleying/luffing (radius r), and hoisting (height z).

Benefits corresponding to this topic:

Can handle the height difference from A→B (B is high), because the lifting mechanism is inherently vertical displacement

Ability to "accurately retrieve the ball"

Comply with input restrictions: the hoisting, turning and moving of the crane can all be achieved by the rotation input through the gear box

B. How to achieve the design goals of this topic

Across 1.2 m and to the higher B: the r, z of the tower crane can naturally handle the span and height difference.

Only rotational input is allowed, and the input shaft is only subject to torque:

The rotation can be driven by gears/friction wheels;

The trolley can use a reel to retract and unwind the rope;

The hoist (drum + reduction) can be used to drive the cable pulley block for lifting.

These are all "rotation inputs"

(2). Billiard ball pickup device (ball pickup mechanism)[2]



Figure 2. Billiard ball picker[3][4]

A. Design review and working principle

Funnel-type guidance: It has a guidance function and reduces the difficulty of alignment.

One-way valve: The ball can go in but not out, to avoid falling when lifting/moving.

Compliance: Use elastic sheets, bristles or flexible components for the entrance, allowing dimensional tolerances and angle errors.

The biggest pain points encountered when picking up balls in box A: the balls are scattered, cannot be sorted by hand, the balls have two weights and have different surface friction characteristics, all of which can be successfully solved by the billiard ball picker.

Guide port (entrance end in box A)

B. How to achieve the design goals of this topic

Do not touch the ball/platform: The ball pick-up entrance can "passively guide" the ball through mechanical guidance without touching the ball with your hands.

Rotation input limitation: The ball-retrieving action can be converted from rotation to straight up and down swing (such as gear and rack)

Success rate of retrieving the ball: The success rate of retrieving the ball is more important than speed, because the ball does not count when it lands. The ball pick-up anti-fall load can reduce the risk of falling.

(3). Natural slope slide (conveyor system)



Figure 3. Natural slope slide[5]

A. Design review and working principle

The slide is a transportation method that "uses gravity as the source of power": as long as the height difference is sufficient, the ball will roll stably to the target in a restricted orbit.

Advantages:

The conveying path can be completely enclosed: reducing the risk of balls falling.

The freedom of movement is restricted: the ball can only move along the track and will not run around and hit the box mouth.

No additional power input is required: the rotation input can be concentrated on "getting the ball + positioning + release", making the whole thing more streamlined and robust.

1.3 General design review

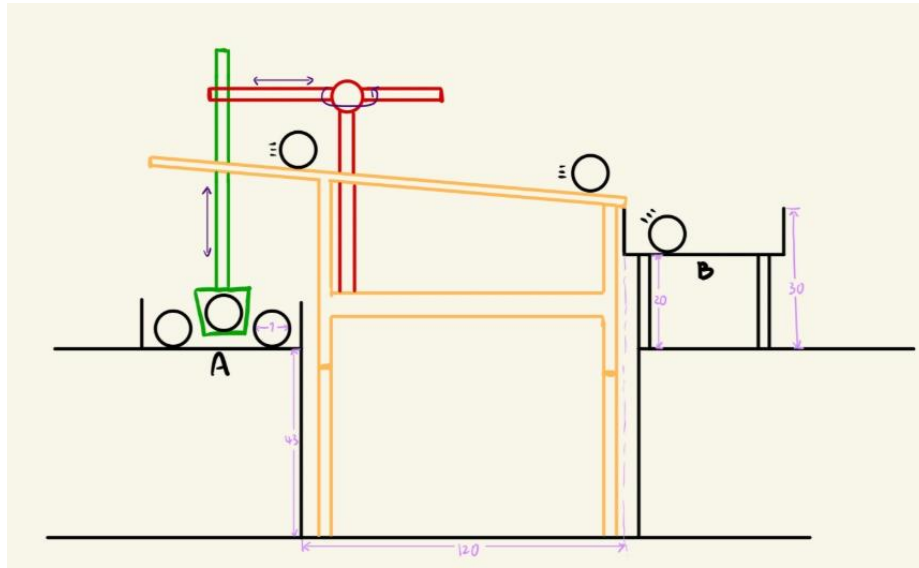


Figure 4. Preliminary general design drawing

This topic is composed of three parts: the ball retrieval module, the polar coordinate positioning module, and the slide transport module. It completes the three functions of ball retrieval, transmission, and ball placement in sequence, and maximizes the number of effective balls in the B box within the specifications.

Ball retrieval module (snooker ball picker concept): The introduction port and one-way valve structure allow the ball to be passively introduced without touching the ball by hand, and the "single ball chamber" ensures that only one ball is picked up at a time, reducing the risk of multi-ball violations and falling.

Positioning module (tower crane concept): Use the column rotation angle θ , arm length radius r and height z to position the end point of the load-bearing ball from side A to the slide entrance to complete the span distance and height difference requirements, and can be realized by the rotation input system.

Transport module (slope slide concept): After the ball is released at the correct position, it rolls to B along the closed slide with gravity. The deceleration and introduction section are designed at the end to avoid losing points due to bouncing to the ground.

Chapter 2, Design Ideas: Introduction to system functional architecture, design

ideas, and performance evaluation

2.1 Design ideas and scoring criteria

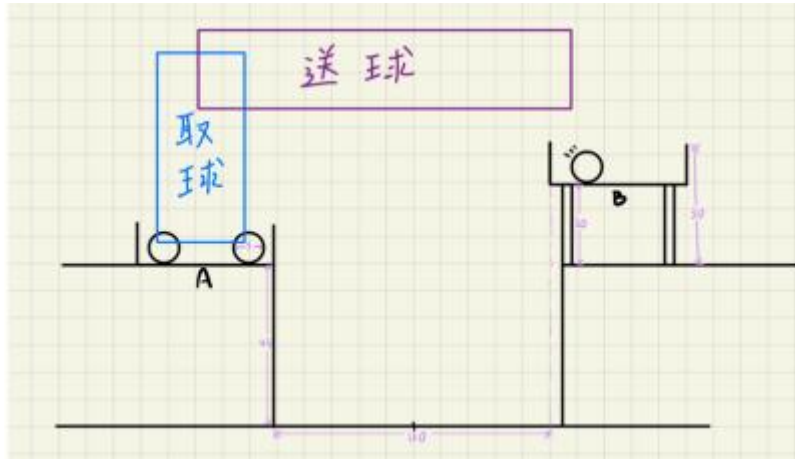


Figure 5. Concept diagram of design ideas

In the initial design idea, we disassembled the overall mechanism into two main bodies, namely the ball-taking and the ball-delivering bodies, as shown in the figure above, and we proposed two design ideas for this concept, namely design one as shown on the left in the figure below and design two as shown on the right in the figure below.

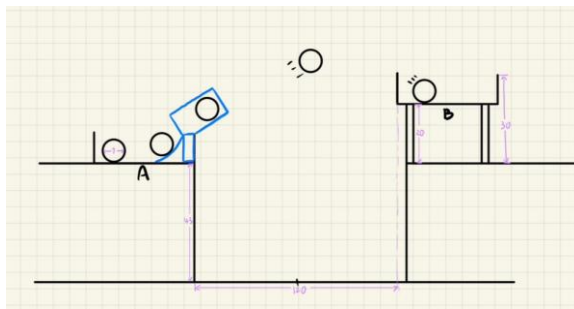
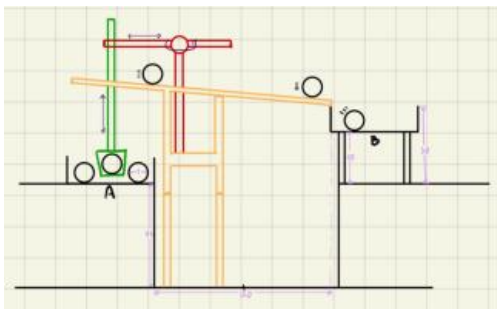


Figure 6: Design Idea 1 Figure 7: Design Idea 2

Both designs have their own individual advantages and disadvantages, so when selecting designs, we customized a set of scoring criteria, scored the overall designs of the two institutions separately, and added them together after weighting, and selected the design with the higher score as our final decision. The following is the scoring table and weighting standards we designed. We scored based on five aspects: creativity, cost, reliability, organization size, and processing convenience:

	weight	design 1	design 2
creativity	1	3	9
cost	3	9	4
reliability	5	8	4
size	3	3	7
machinability	4	8	5
Total		111	82

Table 1. Design idea scoring table[6]

It can be found that after totaling, Design 1 (left in the picture above) has a higher score, so we selected Design 1 as our design idea for this topic.

2.2 Overall design considerations

As described in Chapter 1, our structure mainly consists of three parts (ball retrieval module, polar coordinate positioning module, and slide transportation module). During the design process, we mainly considered the following four aspects, namely foldability, stability, realizability and ease of operation:

Foldability:

The structure should not be too complicated when designing, and the foldable parts and folding methods need to be considered to comply with the storage size (60cm * 60cm * 60cm) stipulated by the rules.

Stability:

It mainly focuses on the selection of materials (dense board, 3D printing, metal) under different stress levels, and the fixing method after deployment. It needs to be able to be firmly fixed while retaining the degree of freedom.

Achievability:

It is necessary to carefully analyze the forces borne by the whole or individual structures when picking up/dribbling the ball to ensure that the structure can operate normally during use.

Ease of operation:

In this timing task, the operator's familiarity with the overall mechanism and the speed of

operation will also be a major focus. The convenience of the operator's operation needs to be added into the design considerations to increase the smoothness and speed of the operation.

2.3 Individual institutional design

(1). Polar coordinate positioning system

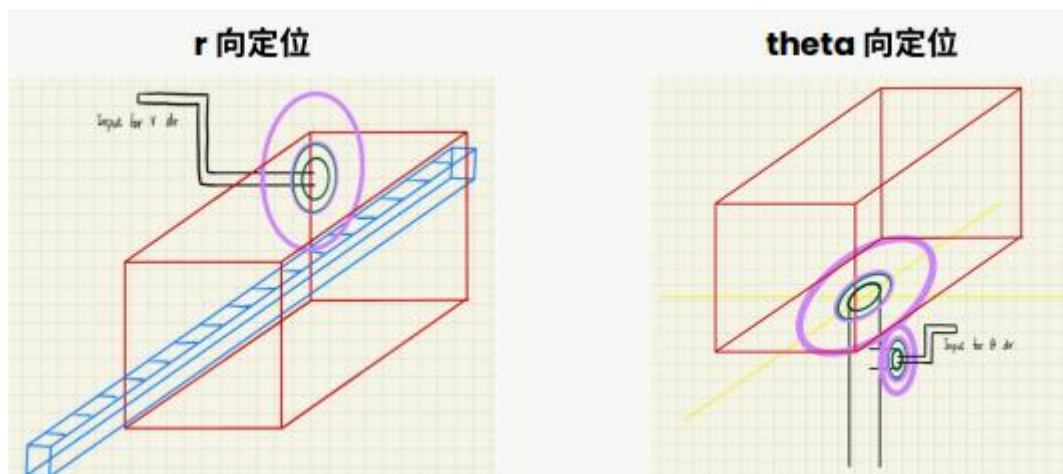


Figure 8. Schematic diagram of branch organization

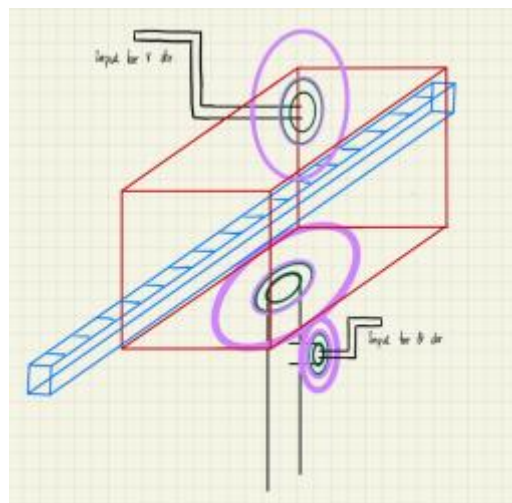


Figure 9. Schematic diagram of the overall mechanism

This design refers to the positioning mode of the tower crane, which requires the two input terminals of r-direction and theta-direction to complete the complete positioning process.

Design considerations:

Gear ratios can be calculated and designed to speed up positioning, and if necessary, torque can be sacrificed in exchange for movement speed

The stability of the structure needs to be carefully considered so that the mechanism does not shake when moving to increase the accuracy of positioning.

This structure will need to support huge positive forces and moments, and careful consideration needs to be given to the stability of the design and the selection of materials.

After analyzing the required size of the mechanism, it was found that in order to be able to pick up the balls from each corner of the box, the length of the boom must be at least $60\sqrt{2}$ cm, which would exceed the specified 60cm, so the folding of the boom needs to be considered.

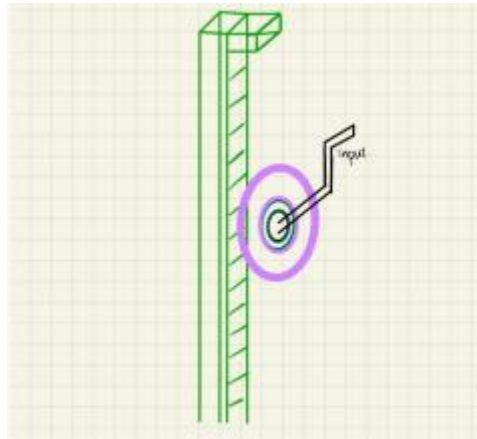
(2). Ball taker-z-direction moving structure

Figure 10. Schematic diagram of z-direction movement structure

This mechanism is only designed for a simple gear to drive the rack to move up and down, and requires an input end to drive the gear.

Design considerations:

And if the input rotation period and gear module are fixed, a gear with a larger diameter can be used to increase the rising and falling speed of the ball taker and improve the speed during operation.

The stability of the structure when rising and falling is also a major consideration. It must be ensured that the mechanism does not shake significantly during operation, which affects the accuracy of positioning.

(3). Ball pick-up box below

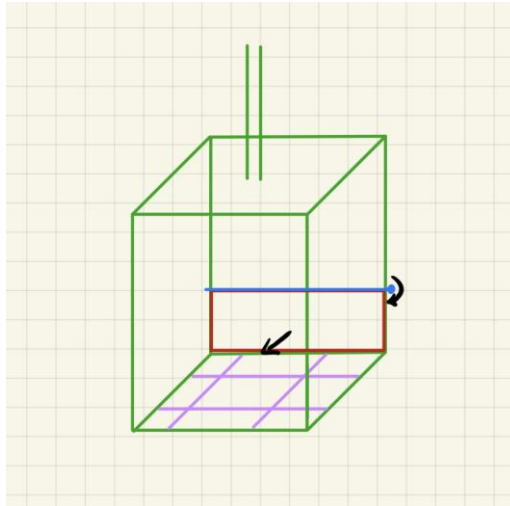


Figure 11. Diagram of taking out the ball box

The design of the ball box is shown in the picture above. We refer to the billiard ball picker as our design template, so that the ball can automatically enter the container without additional input and will not fall out. A rotary input handle needs to be set behind the box opening to help the ball be knocked out of the container and into the track.

Design considerations:

The rotating handle should not protrude too much from the container, as this will affect the ability to pick up the ball at the edge of box A.

The position and angle of the batting board need to be strictly calculated so that the ball can be smoothly hit out of the container without affecting the ball-retrieving function.

The tension of the bottom elastic rope needs to be calculated in detail. The ball must be able to smoothly open the rope and enter the container. When the container rises, the tension must be enough to carry the ball upward.

The weight of the overall ball retriever should not be too heavy to avoid excessive torque when the boom is extended to the farthest end.

(4). Conveying system

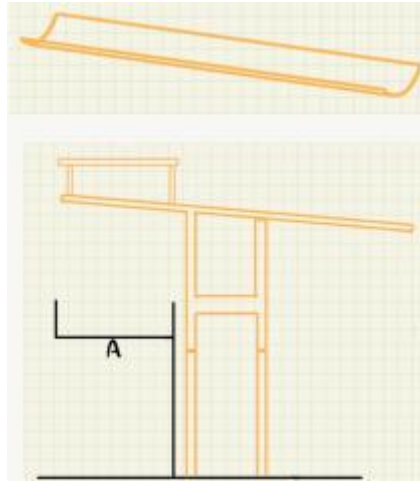


Figure 12. Schematic diagram of conveying system

The conveyor system, as shown in the picture above, is only a simple track system, and is designed to be high on the left and low on the right with reference to the concept of a slope slide, so that the ball can automatically move from box A to box B without any other input.

Design considerations:

A depression is designed in the middle of the track to guide the ball path and prevent the ball from falling out during transportation.

The slope of the track needs to be calculated. While avoiding the overall height of the mechanism, it is necessary to ensure that there is enough slope for the ball to roll on it. The size of the track will exceed the specified 60cm. After the design is completed, the foldable parts and folding methods must be analyzed.

The overall design is connected to the ground on the side of box A. The overall counterweight needs to be considered during design to avoid tipping during operation.

2.4 Use component table

	Materials/components used
polar positioning system	The overall structure of gears, operating handles, etc.: mainly 3D printed parts Rack: dense board + 3D printing If necessary, bearings can be added to the rotating position of the gear to increase the smoothness of operation.
ball taker	The overall structure of gears, operating handles, ball boxes, etc.: mainly 3D printed parts Rack: dense board + 3D printing Elastic cord: hair tie Fold: 3D print
Conveyor system	Overall structure: dense board Folding part: 3D printing + door hinge + hasp lock
shared materials	AB glue, white glue, quick-drying glue, screws, nuts

Table 2. Used component list

Chapter 3 Entity Design and Calculation:

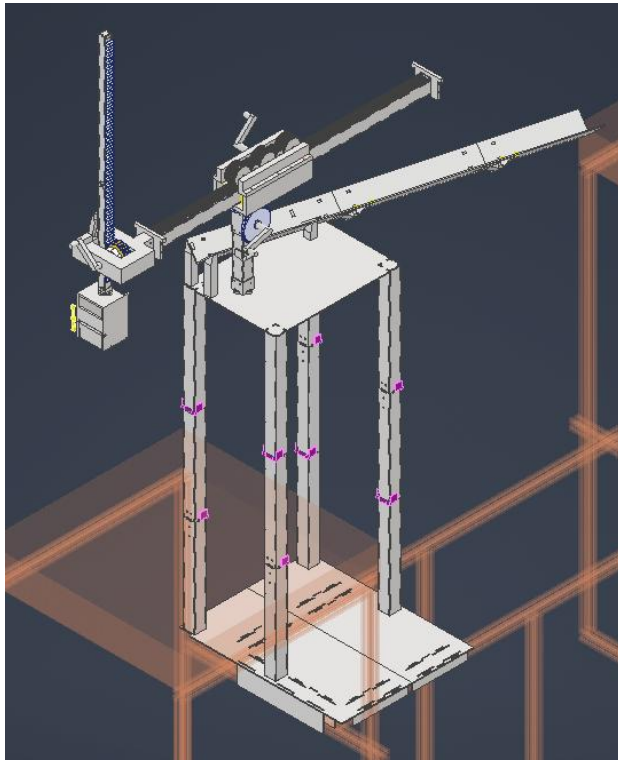
A. Structure introduction

3.1 Overall introduction and configuration (Overview and Configuration)

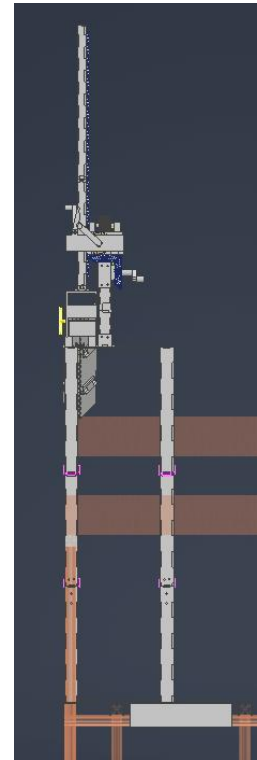
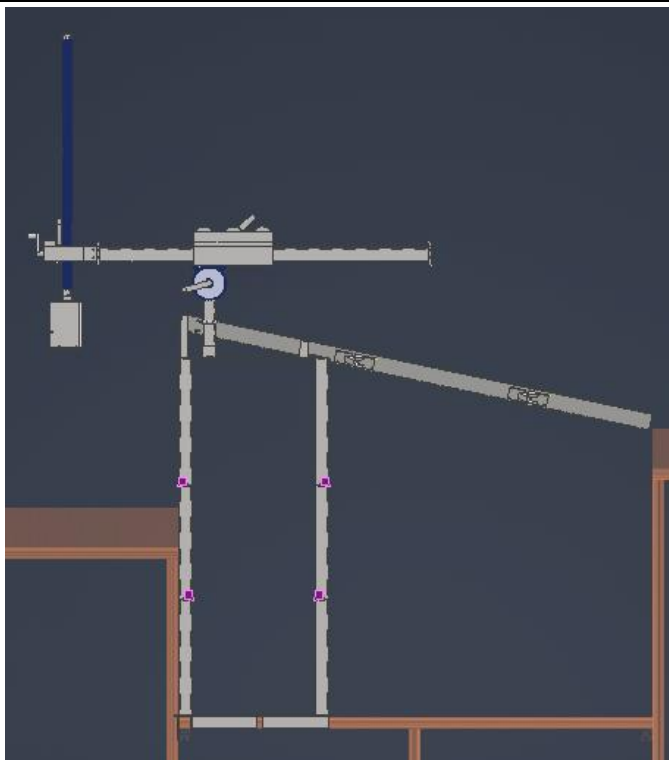
This mechanism is designed to realize automatic grabbing and transportation of spheres. The overall structure adopts the form of a cylindrical coordinate tower crane with three main degrees of freedom (Degree of Freedom, DOF), which respectively control the R, Z, and theta coordinate positions of the gripper in space. The main body of the mechanism consists of a tower crane structure, linear slide rails, rotating base and ball end effector, and is equipped with a gravity track to complete the final transportation.

3.2 General Assembly

overall design	forward view
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side view



looking up

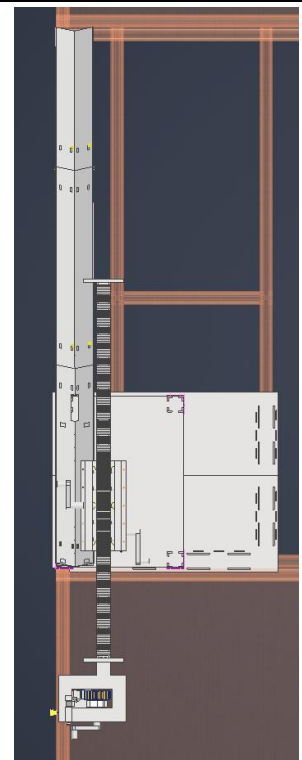


Table 3. Overall combination chart

Note: The engineering drawings are in the appendix. In the selection of engineering drawings, we chose parts that are difficult to see clearly in the general assembly drawing, including ball containers, rack gears, bevel gears, storage shafts, etc.

3.3 Detailed Design

3.3.1 End Effector / Retriever

The ball-retrieval mechanism of this design is divided into two parts: "passive grabbing" and "active unloading":

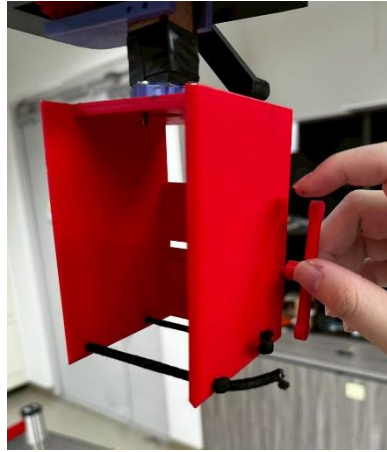


Figure 13. Physical diagram of taking the ball box

Passive capture mechanism (PassiveIntake):

The bottom of the container is equipped with elastic cords (Elastic Cords) arranged in parallel. The design uses the principle of geometric interference to set the distance between the ropes to be slightly smaller than the diameter of the sphere. When the Z-axis moves downward and presses toward the sphere, the sphere is forced to push away the elastic cord and deform. The tension recovery characteristics of the cord are used to buckle the sphere after it enters the container to prevent it from falling.

Active unloading mechanism (ActiveDischarge):

There is a rotating push plate inside the container (driven by a servo motor above the container). When the mechanism moves above the entrance of the track, the push plate rotates, giving the ball a lateral thrust, forcing the ball out of the restraints of the elastic rope and falling into the track.

3.3.2 Transmission and Positioning

The position control core of this mechanism uses a gear and rack (Rack and Pinion) module to convert rotational input into linear output. In view of the different load characteristics of the vertical (Z-axis) and horizontal (R-axis), we have made the following optimized design based on the standard transmission:

1. Vertical lifting mechanism (Z-axis) - ratchet locking and gravity compensation

Since the Z-axis mechanism needs to bear the weight of the ball container and the ball, in order to prevent the mechanism from sliding due to gravity when it is stationary or loses driving force, we integrated a ratchet and pawl (Ratchet and Pawl) device into the transmission system.

Function description:

The ratchet baffle provides one-way mechanical locking function (Self-locking). When adjusted to a specific height, the pawl will snap into the tooth groove to provide reverse support against gravity. This not only avoids the risk of the mechanism accidentally falling, but also eliminates the need for the operator to continuously apply maintenance torque when confirming alignment, greatly improving the stability and safety of the operation.



Figure 14. Vertical lifting mechanism entity

2. Horizontal telescopic mechanism (R-axis) - multi-gear load sharing (Load Sharing)

Considering that the tooth strength (Tooth Strength) of 3D printed gears is limited, if a single gear is used to drive a horizontally extending cantilever structure, the gear teeth may easily break due to stress concentration. Therefore, we adopted a three-gear parallel drive (Parallel Gear Drive) design for the R-axis.

Function description:

Through the three gears meshing with the wide rack at the same time, the total

reaction force during driving is evenly distributed to the three contact points.

According to the principle of mechanical sharing, a single gear only needs to bear one-third of the total load. This design greatly reduces the shear stress at the root of individual gear teeth, effectively solves the problem of tooth chipping due to excessive force during operation, and improves the smoothness of the transmission process.

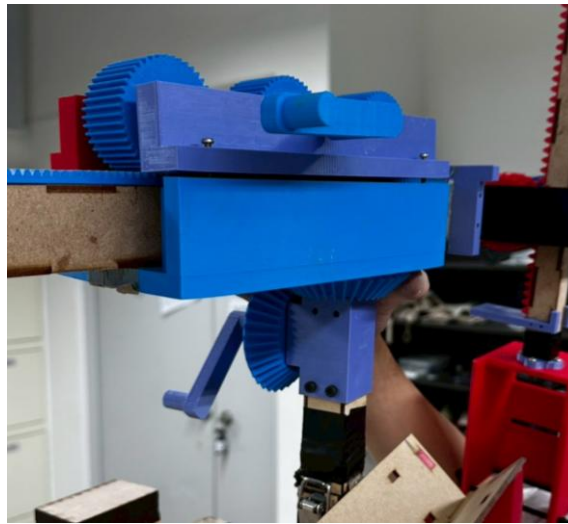


Figure 15. Physical diagram of horizontal telescopic mechanism

3.3.3 Base and Track System

Support base:

It is made of an acrylic plate as the main body, with four support legs underneath, providing enough bottom area to resist the toppling moment generated when the tower crane is extended, ensuring stability during operation.



Figure 16. Physical diagram of the support base

Gravity conveyor track:

The track design exploits potential energy differences for transportation. The track entrance is set within the working range of the tower crane, and a guide funnel is designed to allow for slight position errors during unloading. The track itself has an inclination, and after the ball falls in, it can roll to the end box only by gravity.



Figure 17. Physical diagram of gravity conveyor track

3.3.4 Folding mechanism

This topic has a size limit of 60cm×60cm×60cm for the mechanism, so we made a lot of folding designs to meet the requirements.

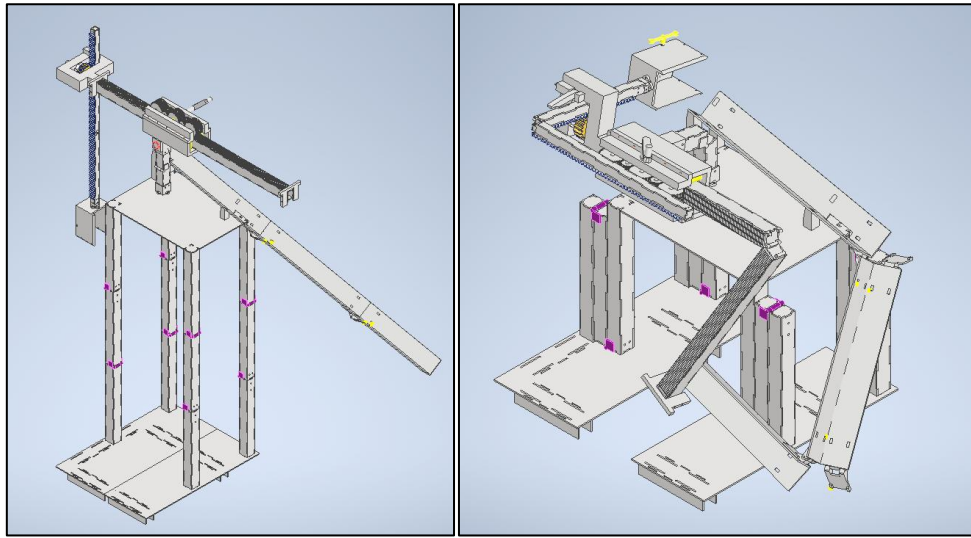


Figure 18. Schematic view after unfolding. Figure 19. Schematic view after folding.

Figure 18 shows the mechanism before folding, and Figure 19 shows the mechanism after folding. It can be placed in a box of 60cm*60cm*60cm.

How it works:

For the rotating mechanism required by the folding mechanism, we use a hinge design to achieve the purpose. At the same time, a buckle lock is added to part of the structure so that the mechanism can be fixed when unfolded.

The base of the tower crane and the supporting feet of the base:

Installing a hinge mechanism on one side of the mechanism allows the device to be folded, and installing hasp locks on the other sides, allowing us to smoothly secure the unfolded mechanism during testing to prevent it from collapsing.

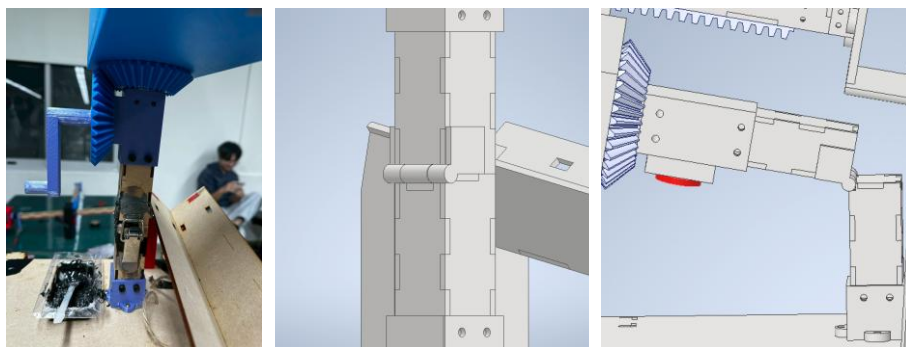


Figure 20, Figure 21, Figure 22, Schematic diagram of the tower crane base folding mechanism

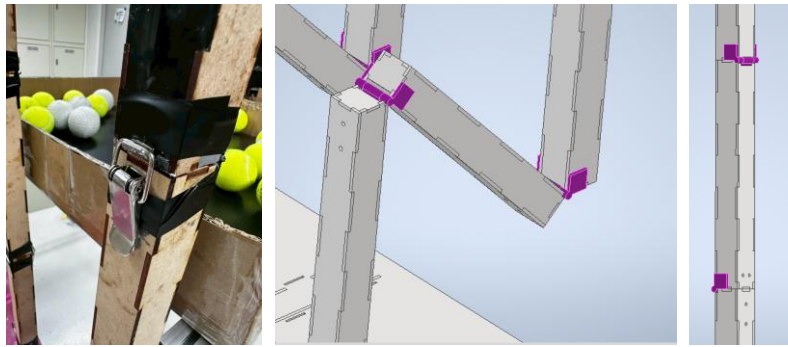


Figure 22, Figure 23, Figure 24. Schematic diagram of the support foot folding mechanism

Vertical lifting (Z-axis) and horizontal telescopic (R-axis):

Install two symmetrical hinge mechanisms at the joint of the mechanism. When folding, only insert a pin into one of the hinges to achieve the purpose of folding and shrinking the mechanism. When you want to unfold the test, insert the other pin to fix the mechanism.

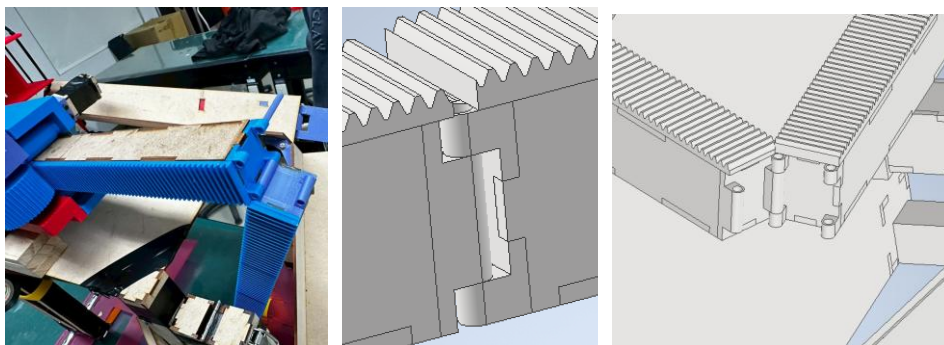


Figure 25, Figure 26, Figure 27. Schematic diagram of horizontal telescopic (R-axis) folding mechanism

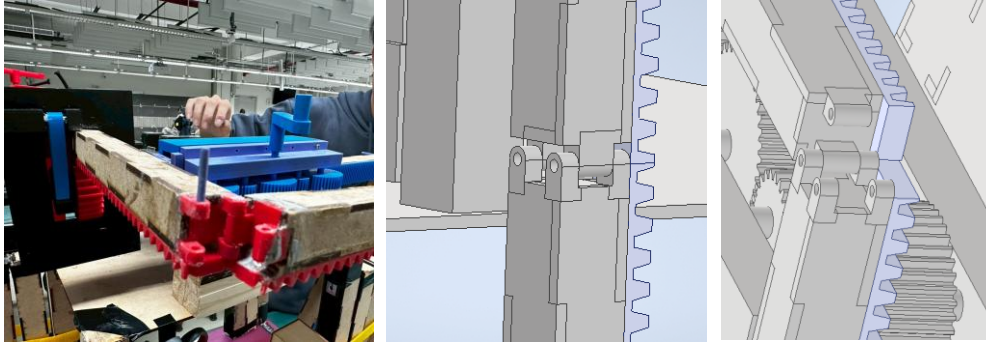


Figure 28, Figure 29, Figure 30, Schematic diagram of the vertical lifting (Z-axis) folding mechanism

Gravity conveyor track:

Install two hinge mechanisms at the joint of the mechanism. When folding, one is only installed on one side, and the other is connected to both sides, so that the mechanism can be folded smoothly. When it is to be unfolded and tested, the other hinge is also connected, and the track can be fixed smoothly after it is stuck.

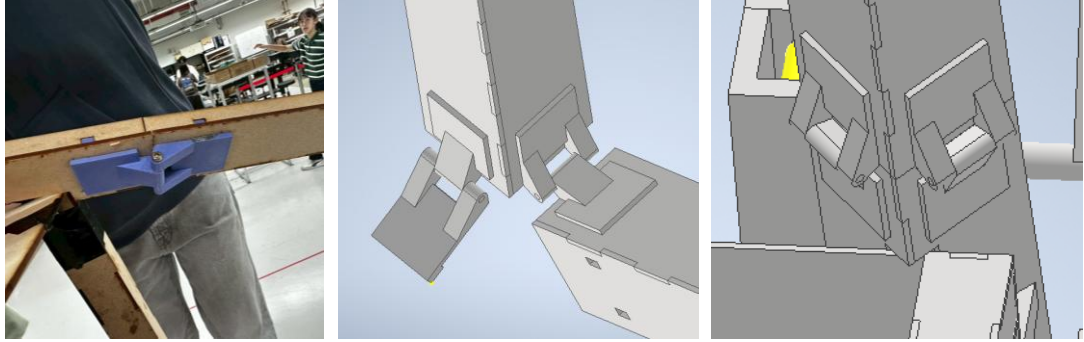


Figure 31, Figure 32, Figure 33. Schematic diagram of the gravity conveyor track folding mechanism

3.4 Operation method and workflow (Operation Sequence)

The complete operation cycle of this organization is as follows:

Zeroing and alignment: The mechanism is started from the initial position, controls the theta axis rotation and adjusts the R-axis elongation so that the ball-taking container accurately stops directly above the target sphere.

Descending ball pickup (Z-axis Down): The Z-axis gear rotates to drive the straight rod to descend, and the elastic rope at the bottom of the container presses the ball to complete passive grabbing.

Lift & Retract: The Z-axis rises to take the ball off the ground, and then the R-axis and theta-axis are linked to transport the ball to the entrance of the track.

Discharge: After confirming that the container is above the track guide opening, start the push plate in the container to push the ball into the track.

Gravity transportation: The ball rolls into the end box along the track, and the mechanism returns to the initial state at the same time, ready for the next grab.

B. Basic calculation[6]

1. Parameters and Assumptions

In order to make the calculation applicable to the input torque and speed, this section adopts the following design parameters (if the actual measurements are different, the values can be updated by substituting the values back into the same set of formulas):

Acceleration due to gravity: $g = 9.81 \text{ m/s}^2$

Ball mass:

Tennis $m_t = 0.058 \text{ kg}$

softball $m_b = 0.111 \text{ kg}$

Equivalent mass of fixture (ball container): $m_g = 0.08 \text{ kg}$

r Equivalent mass of sliding table: $m_c = 0.16 \text{ kg}$

Equivalent radius of drum: $r_d = 0.10 \text{ m}$

Transmission efficiency: conservative (including bearing friction, gear/rope loss) $\eta = 0.8$

r Equivalent kinetic friction coefficient of guide rail (with bearings but still resistance): $\mu_r = 0.05$

Acceleration targets for ball take/placement: $a_z = 1.0 \text{ m/s}^2$

2. Speed calculation

(1). Lifting (Z direction) speed

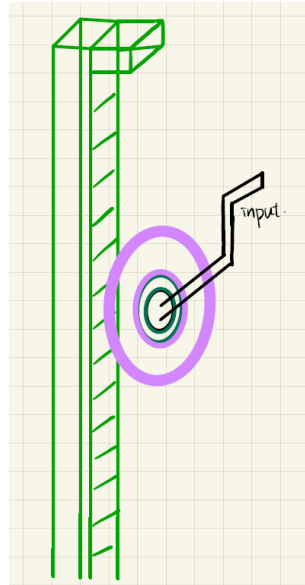


Figure 34. Schematic diagram of z-direction lifting

Assuming a single lifting stroke and using "constant acceleration from stationary lifting"

$$\text{for estimation, then: } s_z = 0.5 m t_z = \sqrt{\frac{2s_z}{a_z}}, \quad v_{z,\text{end}} = a_z t_z, \quad v_{z,\text{avg}} = \frac{s_z}{t_z}$$

$$t_z = \sqrt{1} = 1.0 \text{ s}, \quad v_{z,\text{end}} = 1.0 \text{ m/s}, \quad v_{z,\text{avg}} = 0.5 \text{ m/s}$$

Average drum angular velocity: $\omega_{z,avg} = \frac{v_{z,avg}}{r_d} = \frac{0.5}{0.10} = 5 \text{ rad/s} \approx 47.7 \text{ rpm}$

(2). Radial displacement (r-direction) speed

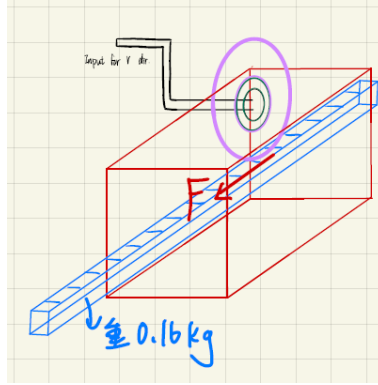


Figure 35. Schematic diagram of r-direction movement

The design goals are that the mining box is 0.6 m wide and can be moved within 1 s. If a more reasonable "symmetrical acceleration and deceleration" is used, the total time

$t_r = 1$ seconds, total displacement m, then the required acceleration: $s_r = 0.6$

$$s_r = \frac{a_r t_r^2}{4} \Rightarrow a_r = \frac{4s_r}{t_r^2} = \frac{4(0.6)}{1^2} = 2.4 \text{ m/s}^2$$

Peak speed vs. average speed: $v_{r,peak} = a_r \frac{t_r}{2} = 2.4(0.5) = 1.2 \text{ m/s}$, $v_{r,avg} = \frac{s_r}{t_r} = 0.6 \text{ m/s}$

Reel angular velocity (average/peak):

$$\omega_{r,avg} = \frac{v_{r,avg}}{r_d} = \frac{0.6}{0.1} = 6 \text{ rad/s} \approx 57.3 \text{ rpm} , \omega_{r,peak} = \frac{v_{r,peak}}{r_d} = \frac{1.2}{0.1} = 12 \text{ rad/s} \approx 114.6 \text{ rpm}$$

(3). Steering (θ direction) speed

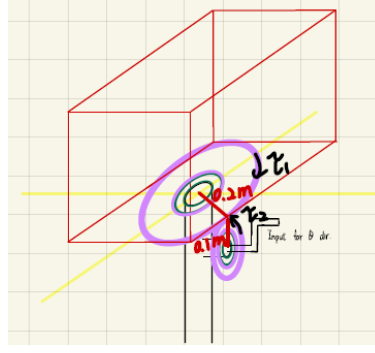


Figure 36: Schematic diagram of steering speed

Suppose it rotates 180° in 2 seconds. Also use symmetrical acceleration and deceleration, total angular displacement rad, time s: $\Delta\theta = \pi t_\theta = 2$

$$\Delta\theta = \frac{\alpha_\theta t_\theta^2}{4} \Rightarrow \alpha_\theta = \frac{4\Delta\theta}{t_\theta^2} = \frac{4\pi}{2^2} = \pi \text{ rad/s}^2$$

Peak angular velocity and average angular velocity:

$$\omega_{\theta, \text{peak}} = \alpha_\theta \frac{t_\theta}{2} = \pi \text{ rad/s} \approx 30 \text{ rpm}, \quad \omega_{\theta, \text{avg}} = \frac{\Delta\theta}{t_\theta} = \frac{\pi}{2} \text{ rad/s} \approx 15 \text{ rpm}$$

(4). Orbital gravity transmission speed

A ball cannot be removed until the previous ball passes the center line (0.6 m). If the transmission adopts inclined rail gravity rolling, the ball is regarded as an approximation of pure rolling without sliding, and its acceleration along the inclined plane is:

$$a = \frac{g \sin \beta - c_{rr} g \cos \beta}{1 + \frac{I}{mR^2}}$$

The ball can be regarded as an approximate solid ball, and we get: $I = \frac{2}{5}mR^2$

$$a = \frac{5}{7}g(\sin \beta - c_{rr} \cos \beta)$$

Take the rolling resistance coefficient, if the design bevel angle is, then $c_{rr} = 0.02\beta = 5^\circ$

$$a \approx \frac{5}{7}(9.81)(\sin 5^\circ - 0.02 \cos 5^\circ) \approx 0.47 \text{ m/s}^2$$

The time it takes for the ball to roll from rest to a distance m from the center line: $s = 0.6$

$$t_{\text{mid}} = \sqrt{\frac{2s}{a}} \approx \sqrt{\frac{1.2}{0.47}} \approx 1.6 \text{ s}$$

Therefore, under this bevel angle setting, the ball can pass the center line in about 1.6 s, which meets the requirement of passing the center line to remove a ball, and the end speed of the ball at a total length of 1.2 m (for evaluation of whether it bounces out of the

ball box too quickly): $v_{\text{end}} = \sqrt{2as} \approx \sqrt{2(0.47)(1.2)} \approx 1.06 \text{ m/s}$

3. Force/torque calculation

(1). Pulling force and input torque required for lifting (Z direction)

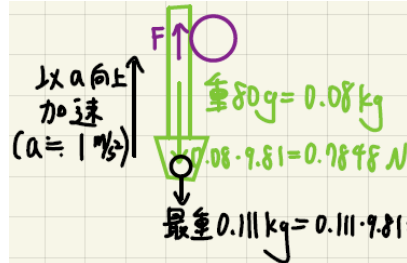


Figure 37: Schematic diagram of stress analysis during the lifting process

It needs to overcome gravity and provide acceleration when lifting. Taking "heaviest ball + clamp" as the worst case:

$$m_{z,\text{max}} = m_g + m_b \Rightarrow F_z = m_{z,\text{max}}(g + a_z)$$

Substitute , , : $m_g = 0.08 \text{ kg}$, $m_b = 0.111 \text{ kg}$, $a_z = 1 \text{ m/s}^2$, $F_z = (0.08 + 0.111)(9.81 + 1) = 2.0647 \text{ N}$

Input torque converted from drum radius (including efficiency):

$$\tau_{z,\text{in}} = \frac{F_z r_d}{\eta} = \frac{(2.0647)(0.10)}{0.8} = 0.258 \text{ N-m}$$

Similarly, if it is a tennis ball (lighter):

$$F_z = (0.08 + 0.058)(10.81) = 1.4918 \text{ N}, \quad \tau_{z,\text{in}} = \frac{1.4918(0.10)}{0.8} = 0.186 \text{ N-m}$$

(2). Radial (r-direction) thrust and input moment

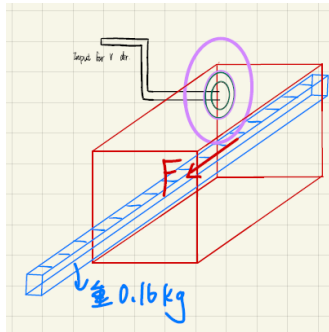


Figure 38: Schematic diagram of force for r-direction movement

When moving in the r direction, acceleration must be provided and guide rail friction must be overcome. Take the total mass of the slide table

$$m_r = m_c + m_g = 0.24 \text{ kg} :$$

$$\text{Substitute , } a_r = 2.4\mu_r = 0.05$$

$$F_r = (0.24)(2.4) + 0.05(0.24)(9.81) = 0.576 + 0.118 = 0.694 \text{ N}$$

Converted drum input torque (including efficiency):

$$\tau_{r,\text{in}} = \frac{F_r r_d}{\eta} = \frac{(0.694)(0.10)}{0.8} = 0.0867 \text{ N-m}$$

(3). Steering (θ direction) moment:

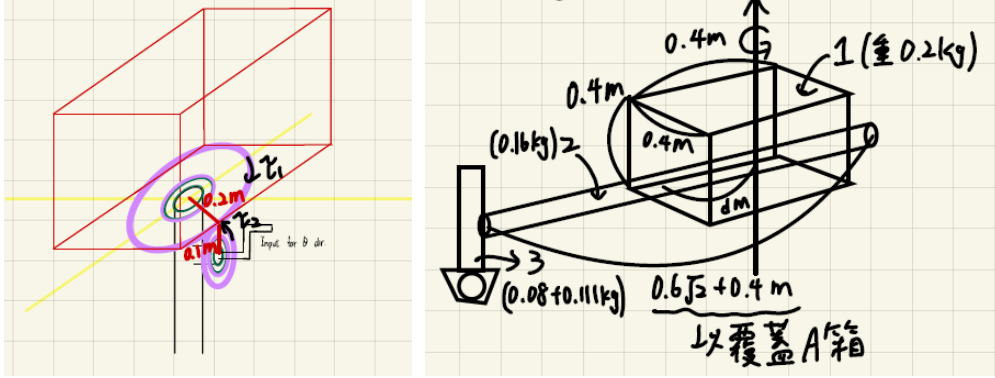


Figure 39 and Figure 40, schematic diagram of force analysis during steering

$$I_1 = \frac{1}{12} 0.2(0.4^2 + 0.4^2) = 0.005333,$$

$$I_2 = \frac{1}{12} 0.16(0.65 + 0.4)^2 + 0.16d^2 = 0.0208 + 0.16d^2,$$

$$I_3 = (0.08 + 0.111)(0.3\sqrt{2} + 0.2 + d)^2 = 0.191d^2 + 0.2385d + 0.0794.$$

$$I_{\text{total}}(d) = 0.35d^2 + 0.238d + 0.15151 \quad (0 \leq d \leq 0.6243 \text{ m})$$

$$\text{Therefore: } I_{\min} = I(0) = 0.15151, \quad I_{\max} = I(0.6243) \approx 0.4365$$

$$\text{From section 2(3). Required torque at output end (main arm): } \alpha_\theta = \pi \tau_{\theta,\text{out}} = I_{\text{total}}(d) \alpha_\theta$$

$$\text{So the range is: } \tau_{\theta,\text{out},\min} = 0.15151\pi = 0.476 \text{ N-m}, \quad \tau_{\theta,\text{out},\max} = 0.4365\pi = 1.37 \text{ N-m}$$

$$\text{If the gear ratio is used, the power relationship is: } i = \omega_{\text{in}} / \omega_{\text{out}} = 2 \tau_{\theta,\text{in}} = \frac{\tau_{\theta,\text{out}}}{\eta i}$$

$$\text{Get: } \tau_{\theta,\text{in},\min} = \frac{0.476}{0.8 \cdot 2} = 0.297 \text{ N-m}, \quad \tau_{\theta,\text{in},\max} = \frac{1.37}{0.8 \cdot 2} = 0.857 \text{ N-m}$$

And the input angular velocity requirement is:

$$\omega_{\theta,\text{in},\text{avg}} = i \omega_{\theta,\text{out},\text{avg}} = 2 \left(\frac{\pi}{2} \right) = \pi \text{ rad/s} \approx 30 \text{ rpm}$$

(4). The clamping force and input torque required to clamp the ball

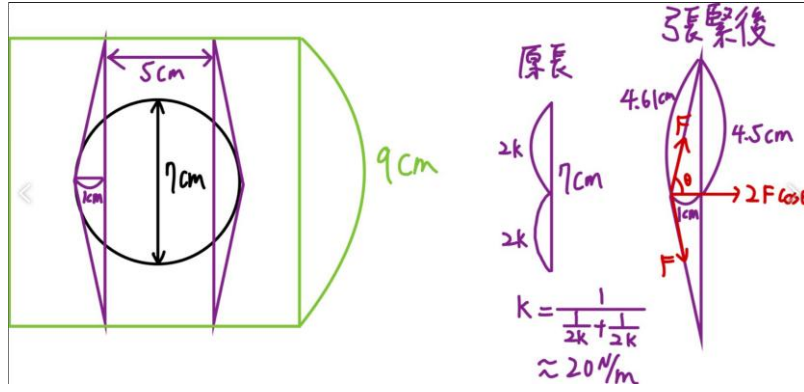


Figure 41: Schematic diagram of force analysis when taking the ball

It is assumed that the clamping ball is provided with friction force by 4 symmetrical contact points (corresponding to the 4 elastic members designed). The normal force at each contact point is N , so the total upward friction available is: $F_{\text{fric, total}} = 4\mu N$

To prevent the ball from falling under the lifting acceleration, it needs to meet: a_z
 $4\mu N \geq m_{\text{ball}}(g + a_z)$

Therefore the normal force required at each contact point is: $N \geq \frac{m_{\text{ball}}(g + a_z)}{4\mu}$

Take the heaviest ball kg, and take the friction coefficient: $m_{\text{ball}} = 0.111$ $a_z = 1$ $\mu = 0.4$

$$N \geq \frac{0.111(9.81 + 1)}{4(0.4)} = \frac{1.200}{1.6} = 0.75 \text{ N}$$

That is, the total normal force is about $N \cdot 4 \approx 3.0$

The normal force is generated by the tension of the elastic member: T

$$N = 2T \cos \theta \Rightarrow T = \frac{N}{2 \cos \theta}$$

$\theta = 77.47^\circ$, so, substitute N : (tension of each elastic member) $\cos \theta \approx 0.2169$ $N = 0.75$

$$T = \frac{0.75}{2(0.2169)} = 1.73 \text{ N}$$

If the elongation of the elastic member is, then linear spring approximation: $\Delta L = 9.22 -$

$$7.00 = 2.22 \text{ cm} = 0.0222 \text{ m} \quad T = k \Delta L \Rightarrow k = \frac{T}{\Delta L} = \frac{1.73}{0.0222} = 78 \text{ N/m}$$

Finally, replace the "ball clamp input" with torque. If the clamping ball drive is equivalent to the drum/cam radius (0.1 m) and needs to overcome the tension of 4 elastic parts: r_c

$$F_{\text{act}} \approx 4T \Rightarrow \tau_{\text{clamp, in}} = \frac{F_{\text{act}} r_c}{\eta} = \frac{(4T) r_c}{\eta}$$

Substitute N , m : $T = 1.73$ $r_c = 0.10$ $\eta = 0.8$

$$\tau_{\text{clamp, in}} = \frac{(4)(1.73)(0.10)}{0.8} = 0.866 \text{ N-m}$$

C.Safety factor calculation[6]

1. Materials and allowable stresses

(1) 3D Print (PLA)

PLA tensile yield strength $S_{y,PLA} \approx 68\text{MPa}$

3D printed parts have problems such as interlayer weakening, filling rate, holes, stress concentration, etc. Use the effective allowable strength:

$$S_{\text{allow},PLA} = 0.30S_{y,PLA} = 0.30 \times 68 = 20.4\text{MPa}$$

(2) Dense board (MDF)

A common range of MDF flexural strength is approximately 23-39 MPa (depending on density): $S_{\text{allow},MDF} = 25\text{MPa}$

(3) AB glue (two-part epoxy)

The strength of lap shear can reach several MPa levels (different base materials vary greatly): $\tau_{\text{allow},epoxy} = 8\text{MPa}$

(4) hinge metal shaft

$$S_{y,steel} = 250\text{MPa}$$

2. Full system load assumptions

(1) Torque can be applied by hand

The average maximum voluntary torque for horizontal operation of the handle is about 4.10 N·m (it will change in different postures) using a slightly conservative design torque:

$$T_d = 4.0 \text{ N-m}$$

(2) Gears, shafts

Gear system (including rack gear, bevel gear)

$$m = 3 \text{ mm}, z = 20, d_p = mz = 60 \text{ mm}, r_p = \frac{d_p}{2} = 30 \text{ mm}$$

$$b = 20 \text{ mm (face width; use } b = 15 \text{ mm)}$$

(3) Pressure angle $\phi=$, $\tan\phi=0.364$ 20°

(4) Lewis form factor $Y=0.33$

(5) Main torsion bearing diameter $d_s=8 \text{ mm}$

(6) The cantilever distance $e=20 \text{ mm}$ from the gear load application point to the nearest bearing

(7) stress concentration

Shaft shoulder/stop screw indentation/keyway, etc.: $K_t = 2.0, K_{ts} = 1.6$

Panel holes/slotted corners (MDF): $K_t = 2.5 \sim 3.0$ (use $K_t =$

2.5 for conservative calculation)

3. Foldable bracket feet (door anklet + AB glue + tower buckle)



Figure 42: Folding bracket entity

(1) Possible stress concentration locations

Near the door hinge hole: The hole causes the cross section to weaken, which is a typical stress concentration.

Tower buckle fixed point: repeated opening and closing will cause local fatigue and hole edge cracks.

AB glue joint edge: Peeling is more likely to break than pure shearing.

(2) Assume that the mechanism + operation causes the maximum bearing capacity of a certain foot:

$$F_{\text{leg}} = 100\text{N}$$

(3) Hinge pin shear stress and safety factor (assuming pin diameter) $d = 4 \text{ mm}$

$$A = \frac{\pi(4)^2}{4} = 12.566\text{mm}^2$$

$$\tau = \frac{F_{\text{leg}}}{A} = \frac{100}{12.566} = 7.96\text{MPa}$$

Shear yield strength of steel $\tau_y \approx 0.577S_y \approx 0.577 \times 250 = 144\text{MPa}$

$$\text{safety factor } n = \frac{\tau_y}{\tau} = \frac{144}{7.96} = 18.1$$

(4) MDF hole edge pressure bearing and safety factor (plate thickness 3 mm)

$$\sigma_b = \frac{F_{\text{leg}}}{td} = \frac{100}{3 \times 4} = 8.34 \text{ MPa}$$

$$\text{safety factor } n = \frac{S_{\text{allow, MDF}}}{\sigma_b} = \frac{25}{8.34} = 3.0$$

(5) AB glue shear stress and safety factor (glued area) $30 \times 50 \text{ mm}$

$$\text{Gluing area } A_g = 30 \times 50 = 1500 \text{ mm}^2$$

$$\text{Rubber shear stress } \tau_g = \frac{F_{\text{leg}}}{A_g} = \frac{100}{1500} = 0.0667 \text{ MPa}$$

$$\text{safety factor } n = \frac{\tau_{\text{allow, epoxy}}}{\tau_g} = \frac{8}{0.0667} = 120$$

4. Bevel gear (miter/bevel) tooth surface meshing stress + gear/shaft fracture safety factor

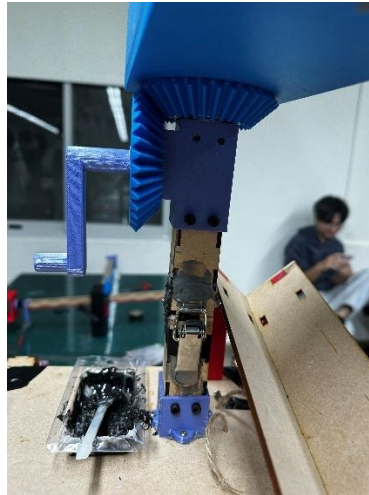


Figure 43: Physical diagram of bevel gear

(1) Bevel gear tangential force (converted from design torque)

$$W_t = \frac{2T_d}{d_p} = \frac{2(4.0)}{0.060} = 133.3 \text{ N}$$

If radial force is required:

$$W_r = W_t \tan \phi = 133.3(0.364) = 48.5 \text{ N}$$

$$W = \sqrt{W_t^2 + W_r^2} = \sqrt{(133.3)^2 + (48.5)^2} = 141.9 \text{ N}$$

(2) Tooth root bending stress (using Lewis approximation, conservatively multiplied by load factor) $K = 1.25$

The bevel gear surface width may be smaller, take:

$$b = 15\text{mm}, \quad m = 3\text{mm}, \quad Y = 0.33, \quad K = 1.25$$

$$\sigma_b = \frac{W_t K}{b m Y} = \frac{133.3(1.25)}{15(3)(0.33)} = \frac{166.7}{14.85} = 11.2\text{MPa}$$

$$n = \frac{S_{\text{allow,PLA}}}{\sigma_b} = \frac{20.4}{11.2} = 1.82$$

(3) Tooth surface contact stress (Hertz contact approximation in AGMA form)

Using the common AGMA contact form (using equivalent elastic coefficient): C_p

$$\sigma_H = C_p \sqrt{\frac{W_t K}{b d_p I}}, \quad C_p = \sqrt{\frac{1}{\pi} \left(\frac{1 - \nu_1^2}{E_1} + \frac{1 - \nu_2^2}{E_2} \right)^{-1}}$$

Take Steel - PLA Contact:

$$E_1 = 200\text{GPa}, \quad \nu_1 = 0.30$$

$$E_2 = 3.5\text{GPa}, \quad \nu_2 = 0.35$$

$$\frac{1 - \nu_1^2}{E_1} = \frac{0.91}{200} = 0.00455\text{GPa}^{-1}, \quad \frac{1 - \nu_2^2}{E_2} = \frac{0.8775}{3.5} = 0.2507\text{GPa}^{-1}$$

$$\Rightarrow \left(\frac{1 - \nu_1^2}{E_1} + \frac{1 - \nu_2^2}{E_2} \right)^{-1} = \frac{1}{0.2553} = 3.918\text{GPa} = 3918\text{MPa}$$

$$C_p = \sqrt{\frac{1}{\pi} (3918)} = \sqrt{1247} = 35.3\sqrt{\text{MPa}}$$

$$\text{Take again } K = 1.25, \quad I = 0.10, \quad b = 15\text{mm}, \quad d_p = 60\text{mm}$$

Contact stress calculation

$$\sigma_H = 35.3 \sqrt{\frac{133.3(1.25)}{15(60)(0.10)}} = 35.3 \sqrt{\frac{166.7}{90}} = 35.3 \sqrt{1.852} = 35.3(1.361) = 48.0\text{MPa}$$

If the compressive yield of PLA is approximately 60 MPa, the "collapse" threshold

$$\Rightarrow n \approx \frac{60}{48.0} = 1.25$$

(4) Resultant stress (including stress concentration) and safety factor of bevel gear shaft

$$\text{Torsional shear stress (solid circular shaft)} \tau = \frac{16T_d}{\pi d_s^3} = \frac{16(4000)}{\pi(8)^3} = \frac{64000}{1608.5} = 39.8\text{MPa}$$

Bending normal stress caused by gear force (in cantilever) $e = 20\text{ mm}$

$$M = We = 141.9(20) = 2838\text{N} \cdot \text{mm}$$

$$\sigma = \frac{32M}{\pi d_s^3} = \frac{32(2838)}{1608.5} = 56.5 \text{ MPa}$$

Equivalent von Mises after stress concentration

$$\sigma' = K_t \sigma = 2.0(56.5) = 113.0 \text{ MPa}$$

$$\tau' = K_{ts} \tau = 1.6(39.8) = 63.7 \text{ MPa}$$

von Mises equivalent stress:

$$\sigma_{vm} = \sqrt{(\sigma')^2 + 3(\tau')^2} = \sqrt{(113.0)^2 + 3(63.7)^2} = \sqrt{12769 + 12173} = \sqrt{24942} = 157.9 \text{ MPa}$$

Safety factor: $n = \frac{S_{y, \text{steel}}}{\sigma_{vm}} = \frac{250}{157.9} = 1.58$

5. Vertical Lift Rack - Gear + Ratchet/Baffle Safety Factor

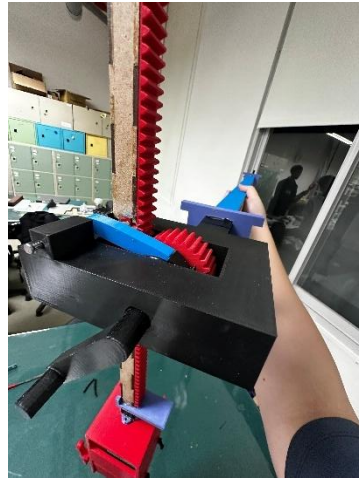


Figure 44: Physical diagram of vertical lifting structure

(1) Vertical lifting load

Assume that the lifting system needs to bear (mechanism's own weight + friction stuck + dynamic operation): $F_z = 30 \text{ N}$

Required gear torque: $T_z = F_z r_p = 30(0.03) = 0.90 \text{ N} \cdot \text{m} = 900 \text{ N} \cdot \text{mm}$

(2) Gear-rack tooth root bending stress and safety factor

$$W_t = F_z = 30 \text{ N}$$

$$\sigma_b = \frac{W_t K}{bmY} = \frac{30(1.25)}{20(3)(0.33)} = \frac{37.5}{19.8} = 1.89 \text{ MPa}$$

$$n = \frac{S_{\text{allow, PLA}}}{\sigma_b} = \frac{20.4}{1.89} = 10.8$$

(3) Lifting shaft (8 mm) resultant stress safety factor

$$\tau = \frac{16T_z}{\pi d_s^3} = \frac{16(900)}{1608.5} = 8.95 \text{ MPa}$$

Gear force (same reason): $W_r = W_t \tan \phi = 30(0.364) = 10.9 \text{ N}$

$$W = \sqrt{W_t^2 + W_r^2} = \sqrt{30^2 + 10.9^2} = 31.9 \text{ N}$$

$$M = We = 31.9(20) = 638 \text{ N} \cdot \text{mm}$$

$$\text{Bending normal stress: } \sigma = \frac{32M}{\pi d_s^3} = \frac{32(638)}{1608.5} = 12.7 \text{ MPa}$$

Contains stress concentration: $\sigma' = K_t \sigma = 2.0(12.7) = 25.4 \text{ MPa}$

$$\tau' = K_{ts} \tau = 1.6(8.95) = 14.3 \text{ MPa}$$

von Mises equivalent stress:

$$\sigma_{vm} = \sqrt{(\sigma')^2 + 3(\tau')^2} = \sqrt{25.4^2 + 3(14.3)^2} = \sqrt{645 + 612} = \sqrt{1257} = 35.5 \text{ MPa}$$

$$n = \frac{S_{y, \text{steel}}}{\sigma_{vm}} = \frac{250}{35.5} = 7.0$$

(4) Ratchet - pawl safety factor

Loaded instantly when stuck $F_p = 100 \text{ N}$

Pawl (PLA) is used as a bending analysis of a small cantilever, assuming pawl cross-

section: $b_p = 20 \text{ mm}$, $t_p = 8 \text{ mm}$, $e_p = 10 \text{ mm}$

$$M_p = F_p e_p = 100(10) = 1000 \text{ N} \cdot \text{mm}$$

$$I_p = \frac{b_p t_p^3}{12} = \frac{20(8)^3}{12} = \frac{20(512)}{12} = 853.3 \text{ mm}^4$$

$$\sigma_p = \frac{M_p c}{I_p} = \frac{1000(4)}{853.3} = 4.69 \text{ MPa}$$

Stress concentration correction (take) $K_t = 2$

$$\sigma_{p, \max} = K_t \sigma_p = 2(4.69) = 9.38 \text{ MPa}$$

$$\text{safety factor } n = \frac{S_{\text{allow, PLA}}}{\sigma_{p, \max}} = \frac{20.4}{9.38} = 2.17$$

6. Calculation of stress concentration and fracture of vertical columns (dense board laser-cut structure + 3D printed door chain support)

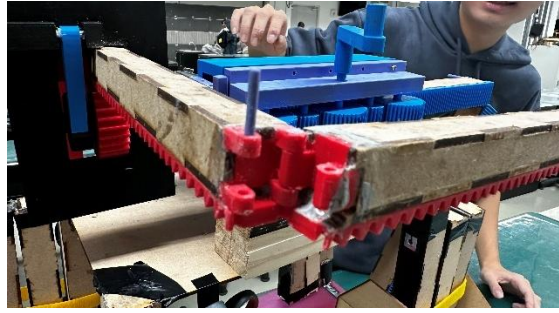


Figure 45: Physical diagram of vertical rack folding

(1) Take the lateral equivalent force: $F_{\text{lat}} = 50\text{N}$

$$L_c = 1.0\text{m} = 1000\text{mm}$$

$$M_c = F_{\text{lat}} L_c = 50(1000) = 50000\text{N} \cdot \text{mm}$$

Equivalent rectangular section ($40 \times 40\text{ mm}$)

Assuming the equivalent section of the column (close to $40 \times 40\text{ mm}$ box shape):

$$I = \frac{bh^3}{12} = \frac{40(40)^3}{12} = 213333\text{mm}^4, c = \frac{h}{2} = 20\text{mm}$$

$$\sigma = \frac{M_c c}{I}$$

Stress concentration (take) $K_t = 2.5$

$$\sigma_{\text{max}} = K_t \sigma = 2.5(4.69) = 11.7\text{MPa}$$

$$\text{safety factor } n = \frac{S_{\text{allow, MDF}}}{\sigma_{\text{max}}} = \frac{25}{11.7} = 2.14$$

7. Safety factor for tower crane system to crossbar (folding) fracture + stress concentration



Figure 46, Figure 47, Figure 48: Physical diagram of boom folding

(1) Bending moment caused by maximum extension

Take the maximum r extension: $r_{\text{max}} = 0.70\text{m}$

End equivalent load (including mechanism self-weight and operation): $F_{\text{end}} = 21\text{N}$

$$M_{\text{arm}} = F_{\text{end}} r_{\text{max}} = 21(0.70) = 14.7\text{N} \cdot \text{m} = 14700\text{N} \cdot \text{mm}$$

(2) Crossbar body bending stress

$$b = 30\text{mm}, \quad h = 60\text{mm}, \quad t = 6\text{mm}$$

$$I = \frac{bh^3 - (b-2t)(h-2t)^3}{12} = \frac{30(60)^3 - 18(48)^3}{12} = \frac{6480000 - 1990656}{12} = 374112\text{mm}^4$$

$$c = \frac{h}{2} = 30\text{mm}$$

$$\sigma = \frac{M_{\text{arm}}c}{I} = \frac{14700(30)}{374112} = 1.18\text{MPa}$$

Stress concentration in holes and slots (take) $K_t = 2.5$

$$\sigma_{\text{max}} = K_t \sigma = 2.5(1.18) = 2.95\text{MPa}$$

$$n = \frac{S_{\text{allow,MDF}}}{\sigma_{\text{max}}} = \frac{25}{2.95} = 8.47$$

(3) Folding hinge hole side pressure bearing

The hinge converts the bending moment into an up-and-down pull pressure couple, equivalent force:

$$F_h = \frac{M_{\text{arm}}}{h} = \frac{14700}{60} = 245\text{N}$$

Assume hinge hole diameter and plate thickness: $d = 5 \text{ mm}$ $t = 6 \text{ mm}$

$$\sigma_{b,h} = \frac{F_h}{td} = \frac{245}{6 \times 5} = 8.17\text{MPa}$$

$$n = \frac{S_{\text{allow,MDF}}}{\sigma_{b,h}} = \frac{25}{8.17} = 3.06$$

8. Bottom plate bracket lateral bending and safety factor

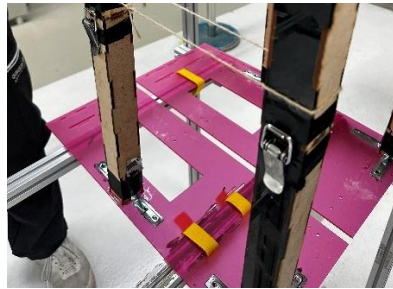


Figure 49: Physical view of the base plate unfolded

Geometry and loading assumptions: $t = 5\text{mm} = 0.005\text{m}$, $b_{\text{eff}} = 0.20\text{m}$, $L = 0.35\text{m}$

The mass of the whole machine is about , take the dynamic load coefficient 2:6 kg
 $W_{\text{total}} = 2mg = 2(6 \times 9.81) = 117.7\text{N} \approx 120\text{N}$

Single side bottom plate load: $P = \frac{W_{\text{total}}}{2} = 60\text{N}$

Maximum bending moment: $M_{\text{max}} = \frac{PL}{4} = \frac{60 \times 0.35}{4} = 5.25\text{N} \cdot \text{m}$

$\sigma_{\text{nom}} = \frac{6M_{\text{max}}}{b_{\text{eff}}t^2} = \frac{6 \times 5.25}{0.20 \times (0.005)^2} = 6.30\text{MPa}$

Stress concentration factor (laser cutting inner angle) $K_t = 1 + 2\sqrt{\frac{a}{\rho}} = 1 + 2\sqrt{5} = 5.47$

$a = 10\text{mm} = 0.010\text{m}$, $\rho = 2\text{mm} = 0.002\text{m}$

Maximum bending stress: $\sigma_{\text{max}} = K_t \sigma_{\text{nom}} = 5.47 \times 6.30 = 34.5\text{MPa}$

Safety factor (based on PMMA yield strength): $S_y = 55\text{MPa}$

$$n = \frac{S_y}{\sigma_{\text{max}}} = \frac{55}{34.5} = 1.59$$

Chapter 4 Experimental Verification and Discussion

A. Experimental verification[6]

1. Verification goals and evaluation indicators

The purpose is to verify the rationality of the material mechanics analysis and safety factor evaluation in the previous chapter. Since moments and stresses were not measured during the implementation process, this study uses measurable physical quantities such as displacement/deflection/angle as the main indicators, and uses structural mechanics equations to derive the equivalent external force and equivalent moment, and then back-calculates the corresponding stress and safety factor.

The verification indicators in this chapter are divided into three categories:

Strength index (stress and safety factor)

Theoretical maximum stress: $\sigma_{\max, \text{theory}}$

Operating actual stress: $\sigma_{\max, \text{actual}}$

Allowable stress (or material strength/allowable value): σ_{allow}

Safety factor:

$$n = \frac{\sigma_{\text{allow}}}{\sigma_{\max}}$$

Stiffness/deformation index (deflection, displacement, angle)

Maximum deflection: $\delta_{\max}$

Maximum lateral displacement: $u_{\max}$

Hinge corner drift: $\Delta\theta$

Functionality and Reliability Index (Performance)

Expand/collapse success rate

Check for looseness, cracks, tooth skipping, or jamming after repeated operations. N

Is the meshing system (gear/rack/ratchet) slipping or reversing failure?

2. Experimental design

Experimental Strategy: Use "Measurable Measurement" to Backward Loading

Since there are no sensors available, the following strategy is adopted:

Static test: Use "known weight (weight)" or "known tension (pulley + weight)" to apply the load, measure the displacement/deflection → reverse the equivalent external force/moment → calculate the stress back.

Dynamic operation test: Actual operation (expanding/folding/rising and descending), use images or displacement measurements to obtain the peak displacement/angle → use the equivalent model to push back the peak value and load → backcalculate the peak stress.

The core concept is: do not measure directly, change the amount, etc., and then use theory to back-reference. $P, T, \sigma, \delta, \theta$

3. Measurement tools and methods

Displacement/deflection measurement

Vernier caliper

Fixed video recording with mobile phone (recommended 60-120 fps) + ruler as background, capturing displacement frame by frame

Angle measurement

Protractor or cell phone angle meter (for hinge angle and swing)

load mode

Weight (known gravity): $P = mg$

Pulley side pull (convert gravity into horizontal force): horizontal pull force is approximate $P \approx mg$

Known moment (weight suspended at known radius): $T = (mg)r$

4. Acrylic base plate lateral bending verification (PMMA)

Equivalent model: The load-bearing belt is regarded as a rectangular cross-section beam (width, thickness, span), and the mid-span is subject to horizontal concentrated force.

Second moment of section: $btLP$

$$I = \frac{bt^3}{12}$$

Mid-span deflection (simply supported beam, mid-span concentrated force):

$$\delta = \frac{PL^3}{48EI}$$

The equivalent external force can be deduced from the measured deflection:

$$P_{eq} = \frac{48EI}{L^3} \delta$$

Maximum bending moment:

$$M_{max} = \frac{P_{eq}L}{4}$$

Nominal bending stress:

$$\sigma_{\text{nom}} = \frac{M_{\text{max}} c}{I}, c = \frac{t}{2}$$

Equivalent simplification (rectangular section):

$$\sigma_{\text{nom}} = \frac{6M_{\text{max}}}{bt^2} = \frac{3P_{\text{eq}}L}{2bt^2}$$

Consider the stress concentration of holes/inner corner gaps:

$$\sigma_{\text{max}} = K_t \sigma_{\text{nom}}$$

Implementation steps (reproducible and accountable)

1. The base is fixed on the tabletop to avoid measurement errors caused by translation of the whole machine.
2. Mark the "maximum bending moment area" of the acrylic sheet (usually the mid-span or mid-point between supports).
3. Use a pulley to convert the gravity of the weight into horizontal tension, and the force application point is fixed at the same position.
4. Set 4-6 loading levels (for example, 10, 20, 30, 40, 50 N), and measure 3 times at each level to average.
5. Use the above formula to infer and calculate back. $P_{\text{eq}} \sigma_{\text{max,actual}}$
6. Take pictures of the edges/inner corners of the holes to see if whitening, micro-cracks, or crack extensions appear to support the determination of stress concentration locations.

Result:

Theoretical maximum stress in the previous chapter: $\sigma_{\text{max,theory}} = 34.5 \text{ MPa}$

Actual maximum stress (scaled to): $P: 60 \rightarrow 40 \text{ N}$

$$\sigma_{\text{max,actual}} = 23.0 \text{ MPa}$$

Error:

$$\text{Signed Error\%} = \frac{23.0 - 34.5}{34.5} \times 100\% = -33.3\%$$

5. Bevel gear: tooth root bending and tooth surface contact

Known torque input (no torque sensor required): Connect a moment arm to the handle or gear shaft and suspend the weight:

$$T_{\text{eq}} = (mg) r$$

Tooth surface tangential force (pitch circle diameter): d

$$F_t = \frac{2T_{\text{eq}}}{d}$$

Tooth root bending stress (same Lewis/AGMA conceptual formula as in the previous chapter):

$$\sigma_F = \frac{F_t}{b m Y}$$

Conservative factors (dynamic loads, load distribution), which can be incorporated into the overall factors: K

$$\sigma_{F, \text{design}} = K \sigma_F$$

Tooth surface contact stress (AGMA contact concept):

$$\sigma_H = C_p \sqrt{\frac{W_t K}{d b I}}$$

Among them. $W_t = F_t$

Implementation steps

1. Install a fixed arm (measuring radius) at the input end. r
2. A series of known torques are obtained by hanging the weight step by step. T_{eq}
3. For each level of torque, operate the transmission 5-10 times and record whether there is slippage, tooth skipping, abnormal noise, and tooth surface wear.
4. Substituting into the above equation to calculate, we get the corresponding $T_{\text{eq}}, \sigma_F, \sigma_H, \sigma_{\text{max, actual}}$

Result:

(1) Bevel gear tooth surface contact

Previous chapter: (with) $\sigma_{H, \text{theory}} = 51.8 \text{ MPa}$ $T = 3 \text{ N-m}$

The relationship between contact stress and loading is approximate, take $\sigma_H \propto \sqrt{F_t} \propto \sqrt{T}$ ($3 \rightarrow 2$) $\sigma_{H, \text{actual}} = 42.3 \text{ MPa}$

Error:

$$\text{Signed Error\%} = \frac{42.3 - 51.8}{51.8} \times 100\% = -18.35\%$$

(2) Bent root of bevel gear

Previous chapter: $\sigma_{F, \text{theory}} = 30.4 \text{ MPa}$

Bending stress is approximated by, take: $\sigma_F \propto F_t \propto T$ ($3 \rightarrow 2$) $\sigma_{F, \text{actual}} = 20.27 \text{ MPa}$

Error:

$$\text{Signed Error\%} = \frac{20.27 - 30.4}{30.4} \times 100\% = -33.3\%$$

6. Input/output shaft: torsion + bending (including stress concentration)

Nominal torsional shear stress:

$$\tau = \frac{16T_{eq}}{\pi d_s^3}$$

Nominal bending normal stress (if the cantilever distance and radial force are known,): $lF_r M = F_r l$

$$\sigma = \frac{32M}{\pi d_s^3}$$

Consider keyway/shoulder etc. stress concentrations (corrected by): K_t, K_{ts}

$$\sigma' = K_t \sigma, \tau' = K_{ts} \tau$$

von Mises equivalent stress:

$$\sigma_{vm} = \sqrt{(\sigma')^2 + 3(\tau')^2}$$

Implementation steps

1. Measured by 5. Bevel gear as shaft torque input. T_{eq}
2. The bending moment is obtained by estimating the gear force model (or replacing it with controlled lateral loading). $F_r M$
3. Calculate back and compare with the previous chapter. $\sigma_{vm,actual} \sigma_{vm,theory}$

(1) Bevel gear steel shaft

Previous chapter: $\sigma_{vm,theory} = 93.2 \text{ MPa}$

Take the same: $T: 3 \rightarrow 2 \sigma_{vm,actual} = 62.1 \text{ MPa}$

Error: Signed Error% = $\frac{62.1-93.2}{93.2} \times 100\% = -33.3\%$

(2) Lifting pinion steel shaft

Previous chapter: (with) $\sigma_{vm,theory} = 72.0 \text{ MPa}$ $F_z = 40 \text{ N}$

$F_z: 40 \rightarrow 30 : \sigma_{vm,actual} = 54.0 \text{ MPa}$

Error: Signed Error% = $\frac{54.0-72.0}{72.0} \times 100\% = -25.0\%$

7. Gear - rack lifting and ratchet baffle

In order to verify the gear shaft stress, pawl root bending stress and anti-fallback ability under lifting load.

The rack bears vertical load (known by hanging weight): $F_z F_z = mg$

$$F_z = mg$$

Gear torque:

$$T = F_z \frac{d_p}{2}$$

The pawl contact force at the effective radius of the ratchet: r_r

$$F_{\text{pawl}} = \frac{T}{r_r}$$

If the impact amplification caused by stuck/instantaneous drop is considered, the dynamic amplification coefficient is taken: K_d

$$F_{\text{pawl,actual}} = K_d F_{\text{pawl}}$$

Bending stress at pawl root (rectangular cross-section, width, thickness, moment arm): bte

$$M = F_{\text{pawl,actual}} e$$

$$Z = \frac{bt^2}{6}$$

$$\sigma_{\text{pawl}} = \frac{M}{Z}$$

If sharp corners/holes in the roots cause concentration:

$$\sigma_{\text{pawl,max}} = K_t \sigma_{\text{pawl}}$$

Implementation steps

1. Hang a known weight (e.g. 1-5 kg) on the lifting platform to obtain a known F_z
2. Operate the ascending/descending operation and trigger the ratchet lock during the descending section to record whether it slips or not.
3. If it slips, record the slip load threshold $F_{z,\text{fail}}$
4. Calculate the maximum stress of the pawl based on the actual locking load, and check whether the prediction in the previous chapter is conservative.

Result:

Previous chapter: $\sigma_{\text{pawl,theory}} = 26.6 \text{ MPa}$

The actual impact is lower than the conservative assumption (equivalent effectiveness ratio): $80 \rightarrow 60 \text{ N}$

$$\sigma_{\text{pawl,actual}} = 19.95 \text{ MPa}$$

Error:

$$\text{Signed Error\%} = \frac{19.95 - 26.6}{26.6} \times 100\% = -25.0\%$$

8. Discussion of error reasons

Comparison shows that the actual stress is generally lower than the theoretical maximum stress in the previous chapter, with an error of about 18% to 33%. This result is reasonable from an engineering perspective, and the reasons can be divided into four categories: "loading uncertainty," "model assumption differences," "manufacturing and material differences," and "measurement and inversion errors."

(1). Loading uncertainty (the main source)

The manual operating torque is not constant: there is a large gap between the short-term peak value and the average value; the previous chapter used a conservative peak value design, while this chapter uses a more reasonable operating average peak value. 3 N-m 2 N-m

Instantaneous dynamic load caused by jamming/collision: If the tooth surface suddenly bites or rebounds, the contact stress will rise instantly, but it is difficult to capture the true peak value with general displacement measurement.

Load shunting: Acrylic plates, aluminum extrusions, wooden boards, 3D printing parts, and screw pre-tightening forces in the entire machine will share the lateral force, causing the actual bearing ratio of acrylic to be lower than the ideal assumption of "all borne by acrylic."

(2). Boundary conditions and theoretical model simplification (systematic errors)

Difference between simply supported/fixed support: Acrylic panels are actually mostly screw-locked and clamped, close to semi-fixed support; if simply supported in theory, the rigidity will usually be underestimated and affect the inversion results.

The beam model ignores the plate effect: the acrylic base plate is a plate rather than a pure beam, and the two-way load bearing of the plate will make the actual stress distribution different from that of a one-dimensional beam.

The gear model ignores eccentricity and tooth surface roughness: the formulas in the previous chapter are mostly ideal meshes; actual printing of tooth surface roughness and eccentricity will cause local contact to be more peaky (this is why tooth surface contact is the most sensitive).

(3). Materials and processes (theoretical strength/concentration coefficient is different from the actual product)

3D printing anisotropy: PLA strength is affected by layer orientation, filling rate, pores, and temperature, and the equivalent value may be low. S_y

Acrylic has high notch sensitivity: Microcracks and heat-affected zones at the inner corners of the laser cut will change the equivalent and crack initiation thresholds. K_t

Hole edge processing quality: burrs and scratches will cause cracks to occur earlier, causing the safety factor to be worse than the theory (or conversely, if chamfering/polishing is provided, it will be better).

(4). Measurement and inference errors (errors in the experimental method itself)

The displacement measurement resolution is insufficient: the deflection is only 0.1-0.3 mm, and there is an error of 10% to 20% in cell phone grid or ruler measurement.

The measuring point is not the true maximum deflection position: causing inverse deviation. P_{eq}

Loading direction deviation: If the horizontal tension force has a vertical component, it will change the force path and stress distribution.

B. Problems encountered and ways to improve

1. Structural Stiffness and Deformation are not considered during design.

Only geometric motion is considered during the design phase, and the mechanical behavior of the material under load is not fully evaluated, resulting in the following unstable conditions:

(1). Cantilever beam effect causes bearing failure

Problem description: The tower crane crossbar is a typical cantilever beam (Cantilever Beam) structure. When the mechanism extends to grasp a distant target, the force arm increases, causing the root to bear a huge bending moment (Bending Moment). This bending moment is transmitted to the rotating bearing. Generally, deep groove ball bearings mainly bear radial force and cannot withstand excessive moment loads, resulting in damage to the bearing structure during testing.

Solution: If the bearing model cannot be changed, use adhesive to fill the gap between the bearing and the seat hole to increase the contact area and disperse the stress to prevent the bearing from falling off or damage caused by stress concentration.

(2). Insufficient rigidity of the base causes overall skew

Problem description: The main mechanism adopts a partially suspended design, relying only on the bottom acrylic plate for support. Due to the insufficient flexural stiffness of the acrylic plate, deflection occurs under load, which also causes the tower crane mechanism above to tilt, affecting positioning accuracy.

Solution: Add stiffener/Rib under the base plate - stick long wooden blocks. This effectively increases the cross-sectional moment of inertia of the base and significantly reduces bending deformation.

(3). Sagging occurs when the track span is too long.

Problem description: The middle and rear sections of the transmission track lack support, and are affected by the dynamic load of its own gravity and ball rolling, causing downward deflection. This resulted in a lower-than-expected height at the end of the track, resulting in insufficient elevation difference between the exit and the receiving box for the sphere's kinetic energy to overcome the potential energy barrier.

Solution: Make end geometry corrections, use tape to raise the track exit, and adjust the slope and connection angle to ensure that the sphere can slide into the box smoothly.

2. Assembly sequence and design for manufacturability (DFA - Design for Assembly)

Problem description: When designing the dense fiberboard (MDF) structure, the access space (Tool Access) of hand tools (wrenches, screwdrivers) was not considered. After gluing the outer panels, it was discovered that the inner nut was sealed and could not be locked.

Solution: Forced to change the combination method and change the original screw lock design to adhesive fixation. Although the assembly problem was solved, it also resulted in the structural strength at the turning point being lower than the original bolt-joint design, and the flexibility for repair and disassembly was lost.

3. Tolerance and Manufacturing Processes

Problem description: Mixing parts of different processes leads to difficulty in matching:

Laser Cutting: The dense board is ablated by laser to produce kerfs, resulting in larger apertures and loose tenons.

3D printing (FDM): The thermoplastic material shrinks after cooling, resulting in smaller pore diameter and larger outer diameter (Elephant foot effect).

When the two fit together (such as a 3D print embedded in an MDF hole), it is easy to produce unexpected excessive interference (Interference Fit) or excessive gaps.

Solution: Post-processing/Sanding the interference areas; use filler or adhesive to fill the gaps that are too large to achieve appropriate fit tolerances.

4. Mechanism and Dynamics

(1). Incompatibility between standard parts and network resources

Problem description: The Retriever component originally used open source gear and rack files on the Internet. After actual printing, it was found that the modules or pressure angles of the two did not match, resulting in the inability to mesh smoothly.

Solution: Give up Internet resources, carry out parametric design according to the needs of the organization, and draw gears and racks that meet the specifications.

(2). Dynamic oscillation caused by tooth backlash

Problem description: There is backlash between the gear and the rack. When the tower crane moves rapidly or changes direction, this gap causes residual oscillation (Oscillation) at the end of the mechanism. Excessive vibration amplitude causes the ball to become unstable during transportation and even slip from the claws, seriously affecting the transmission efficiency.

Chapter 5, Conclusion and Future Outlook

1. Conclusion

This topic successfully designed and implemented a set of polar coordinate positioning crane pick-up and drop-off systems that meet the specifications. The overall mechanism uses "rotation input" as the only operation mode, integrating the ball pick-up module, polar coordinate positioning module and gravity rail transport module to complete the task of picking up the ball from platform A and transporting it to platform B without direct contact with the ball and the platform.

In the mechanism design part, we took the daily engineering mechanism as the starting point and drew on the cylindrical coordinate positioning principle of the tower crane, the passive guidance and one-way anti-drop mechanism of the table tennis ball picker, and the gravity transportation characteristics of the slope slide. We successfully transformed the mature and stable engineering concept into a special mechanism design that meets the course constraints. This design strategy effectively reduces the system complexity and improves the reliability and stability of the overall operation.

The overall system has three main degrees of freedom (R , Z , θ), and can convert rotational input into linear and angular motion through gears, racks, bevel gears and drum mechanisms to achieve precise positioning. In order to meet the structural load-bearing and safety requirements, a number of key designs have been added to the system, including:

The ratchet-pawl self-locking mechanism of the Z -axis lifting system effectively prevents gravity from falling back.

The R -axis adopts a multi-gear parallel load sharing design to reduce the tooth root stress of a single gear.

The bottom of the θ rotating shaft (at the bevel gear) uses bearings to bear the radial force generated by the mechanism and reduce the friction generated during rotation.

The base and column structure are combined with support legs and a folding mechanism to take into account stability and volume constraints.

The ball retrieval container at the end is passively clamped by elastic parts to ensure single ball retrieval and prevent falling.

Regarding theoretical analysis and calculation, we established a complete speed, force and

torque estimation model for the lifting, radial movement, rotation positioning and ball clamping mechanism, and further conducted material strength and safety factor analysis. The calculation results show that under conservative assumptions and the most unfavorable load conditions, all key components (gears, shafts, ratchets, bottom plates, folding hinges, etc.) have sufficient safety factors and can withstand the load required for actual operation.

In terms of experimental verification and test results, this study indirectly verified the rationality of the aforementioned theoretical analysis by using the method of "measurable displacement inverse equivalent loading". The results show that the actual measured equivalent stress is generally lower than the theoretical maximum stress, and the error falls within the engineering acceptable range of about 18% to 33%, indicating that the previous design calculation has a conservative but reasonable safety margin. In addition, repeated operation tests also confirmed that the system did not suffer serious structural failure after multiple deployments, collapses and operations, and it has a certain degree of reliability.

However, during the process we also found that structural rigidity, dynamic stability and manufacturability have a significant impact on overall performance in actual operation. Some designs did not fully consider mechanics and assembly in the early stages, resulting in problems that had to be solved through reinforcement, correction and compromise during the implementation stage. These experiences highlight the importance of simultaneously integrating geometric feasibility, mechanical analysis and manufacturing constraints into machine design.

2. Future Work

We mentioned many problems we encountered during the implementation process. Based on these experiences, we believe that we can extend and improve from the following aspects in the future:

(1). Structural rigidity and material selection optimization

Change the main load-bearing components from MDF or acrylic to aluminum extrusions or thin-walled steel structures (wood can also be used if there are budget constraints) to significantly improve the overall bending and torsional rigidity.

For the cantilever structure, a triangular brace or frame design is introduced to reduce the problem of moment concentration on the rotating bearing.

Redesign the base into a box or rib structure to increase the section moment of inertia and reduce deflection.

(2). Improvement of transmission system and dynamic stability

Improve the backlash problem between gears and racks, such as adding an elastic preload mechanism or a double rack anti-backlash design to reduce vibration when switching directions.

Evaluate the use of ball screws or synchronous belts to replace part of the gear transmission to improve smoothness and positioning accuracy.

(3). Folding mechanism and durability design

Redesign the folding hinge and latch structure into special joints to avoid stress concentration and fatigue damage.

Evaluate the fatigue life under the number of folding and unfolding cycles, and introduce higher-durability metal joints or quick-release structures.

(4). Improvement of ball retrieval mechanism

Carry out more systematic elastic element selection and parameter design for the elastic clamping mechanism, so that it can respond more stably to different ball weights and surface friction differences.

(5). Control and automation extension

If you are not limited by "pure rotation input", you can introduce servo motors and control systems to achieve semi-automatic or fully automatic operation.

Combined with a position sensor or a simple visual aid system to improve ball picking and

positioning accuracy.

Further explore the optimization of the shortest path and operating rhythm to improve overall operating efficiency.

(6). Extended learning of engineering design process

In the future, finite element analysis (FEA) can be combined at the early stage of design to predict high-stress areas in advance and reduce the need for corrections after implementation.

Incorporate DFA/DFM (Design for Assembly/Manufacturing) principles into the design stage to reduce assembly difficulty and rework risks.

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Appendix

